

Calculating Zoro's Bite Force

A Physics-Based Analysis of the Three-Sword Style Cannon Scene

Abstract

In *One Piece*, Roronoa Zoro uses the Three-Sword Style by holding one sword in each hand and a third sword in his mouth. This unusual fighting style raises an interesting physics question: how much jaw clamping force would theoretically be required to keep the sword in position during an extreme impact?

This study develops a simplified mechanical model of the mouth-held sword and estimates the jaw clamping force required during a modeled cannonball impact.

The sword is modeled as a rigid object with a mass of 1.5 kg and a length of 1.35 m. The center of mass and the modeled impact point are both placed 0.40 m from the jaw. The effective jaw contact distance is assumed to be 0.05 m.

The cannonball is modeled with a mass of 15.0 kg, a velocity of 450 m/s, and an assumed collision duration of 0.015 s.

Under the maximum-momentum-transfer assumption, the cannonball has a momentum of

$$p = 6750 \text{ kg m/s.}$$

Dividing this momentum change by the assumed collision time gives an average impact force of

$$F_{\text{impact}} = 450000 \text{ N.}$$

At an impact distance of 0.40 m from the jaw, this corresponds to an impact torque of

$$\tau_{\text{impact}} = 180000 \text{ N m.}$$

Using the simplified jaw torque model

$$\tau_{\text{resist}} = F_{\text{jaw}} d_{\text{contact}},$$

the corresponding theoretical jaw force is

$$F_{\text{jaw}} = 3600000 \text{ N} \approx 367.10 \text{ tf.}$$

This value is not a measurement of Zoro’s biological bite force. It is the force required by the assumptions of this simplified mechanical model.

The accompanying Python simulation uses the same primary constants and identifies approximately the same threshold magnitude under its own numerical success criteria.

1 Motivation

One Piece has been loved by people around the world for many years, and I am one of those fans.

One of the most recognizable aspects of Roronoa Zoro’s fighting style is the Three-Sword Style. Zoro holds one sword in each hand while gripping a third sword with his mouth.

This creates an unusual mechanical problem.

The mouth-held sword is not simply held stationary. Zoro moves, changes direction, attacks, blocks, and experiences impacts while maintaining control of the sword.

This leads to the following question:

How much jaw clamping force would be required to keep a sword secured under an extreme mechanical impact?

The purpose of this research is not to determine an official value for Zoro’s bite force.

Instead, the goal is to construct a simplified physics model and determine what magnitude of jaw force would be required if the fictional scene were approximated using classical mechanics.

2 Research Question

The main research question is:

Under the assumptions of a simplified rigid-body model, what jaw clamping force would be required for Zoro to resist the modeled cannonball impact?

A secondary goal is to compare the analytical calculation with the output of the Python simulation used for this experiment.

3 Model Overview

The model follows the following sequence:

1. Define the sword mass and geometry.

2. Determine the gravitational force acting on the sword.
3. Calculate the gravitational torque about the jaw.
4. Determine the ideal jaw force required for static horizontal equilibrium.
5. Define the cannonball mass and velocity.
6. Calculate the cannonball momentum.
7. Estimate the average impact force using the assumed collision duration.
8. Convert the impact force into a torque about the jaw.
9. Calculate the jaw force required to provide an equal resisting torque.
10. Evaluate the simplified friction condition.
11. Compare the analytical result with the Python simulation.
12. Examine the sensitivity of the result to major assumptions.

4 Model Constants

The primary constants used by the Python simulation are reproduced below.

Table 1: Primary model constants

Parameter	Value
Gravitational acceleration, g	9.80665 m/s ²
Sword mass	1.5 kg
Sword length	1.35 m
Center-of-mass distance from jaw	0.40 m
Modeled impact distance from jaw	0.40 m
Jaw contact distance	0.05 m
Friction coefficient, μ	0.5
Cannonball mass	15.0 kg
Cannonball velocity	450 m/s
Assumed collision duration	0.015 s
Maximum allowed rotation	5.0°
Maximum modeled jaw opening	5.1 cm

The model therefore uses the following primary geometric quantities:

$$r_{\text{com}} = 0.40 \text{ m}$$

$$r_{\text{impact}} = 0.40 \text{ m}$$

and

$$d_{\text{contact}} = 0.05 \text{ m}.$$

These distances are especially important because torque depends directly on the lever arm.

5 Unit Conversions

The calculations are performed internally using SI units.

The relationship between kilograms-force and newtons is

$$1 \text{ kgf} = 9.80665 \text{ N}. \quad (1)$$

One metric ton-force is defined as

$$1 \text{ tf} = 1000 \text{ kgf}. \quad (2)$$

Therefore,

$$1 \text{ tf} = 1000 \times 9.80665 \quad (3)$$

and hence

$$\boxed{1 \text{ tf} = 9806.65 \text{ N}.} \quad (4)$$

The reverse conversion is

$$F_{\text{tf}} = \frac{F_{\text{N}}}{9806.65}. \quad (5)$$

This conversion is used throughout the paper.

6 Sword Weight

The gravitational force acting on the sword is

$$W_s = m_s g. \quad (6)$$

Using

$$m_s = 1.5 \text{ kg}$$

and

$$g = 9.80665 \text{ m/s}^2,$$

we obtain

$$W_s = 1.5 \times 9.80665. \quad (7)$$

Therefore,

$$\boxed{W_s = 14.709975 \text{ N.}} \quad (8)$$

Rounded appropriately,

$$W_s \approx 14.71 \text{ N.}$$

7 Static Gravitational Torque

The sword's weight acts at its assumed center of mass.

The gravitational torque about the jaw is

$$\tau_g = W_s r_{\text{com}}. \quad (9)$$

Substituting,

$$\tau_g = 14.709975 \times 0.40. \quad (10)$$

Therefore,

$$\boxed{\tau_g = 5.88399 \text{ N m.}} \quad (11)$$

Thus the sword produces only approximately

$$5.884 \text{ N m}$$

of gravitational torque under the assumptions of the model.

8 Jaw Force Required for Static Equilibrium

The simplified model assumes that jaw clamping force produces a resisting torque according to

$$\tau_{\text{resist}} = F_{\text{jaw}} d_{\text{contact}}. \quad (12)$$

For static equilibrium,

$$\tau_{\text{resist}} \geq \tau_g. \quad (13)$$

Therefore,

$$F_{\text{jaw}} \geq \frac{\tau_g}{d_{\text{contact}}}. \quad (14)$$

Substituting,

$$F_{\text{jaw}} \geq \frac{5.88399}{0.05}. \quad (15)$$

This gives

$$F_{\text{jaw}} \geq 117.6798 \text{ N}. \quad (16)$$

Converting to kilograms-force:

$$F_{\text{jaw}} = \frac{117.6798}{9.80665}. \quad (17)$$

Therefore,

$$\boxed{F_{\text{jaw,static}} = 12.00 \text{ kgf.}} \quad (18)$$

This is the idealized torque-balance requirement for simply holding the sword horizontally.

It is important to note that this result is dramatically smaller than the force required for the modeled cannonball impact.

9 External Force Example

Torque depends not only on force but also on the distance from the pivot:

$$\tau = Fr. \quad (19)$$

Consider an external force of

$$F_{\text{external}} = 3 \text{ kgf.}$$

Converting:

$$F_{\text{external}} = 3 \times 9.80665 = 29.41995 \text{ N.} \quad (20)$$

If this force acts at

$$r = 0.40 \text{ m,}$$

then

$$\tau_{\text{external}} = 29.41995 \times 0.40. \quad (21)$$

Therefore,

$$\boxed{\tau_{\text{external}} = 11.76798 \text{ N m.}} \quad (22)$$

This is approximately twice the sword's gravitational torque.

The example demonstrates why an impact far from the jaw can become mechanically significant even when the force itself is not enormous.

10 Cannonball Impact Model

The extreme scenario uses the following assumptions:

$$m_c = 15.0 \text{ kg}$$

$$v_c = 450 \text{ m/s}$$

$$\Delta t = 0.015 \text{ s.}$$

The cannonball is assumed to interact with the sword at

$$r_{\text{impact}} = 0.40 \text{ m}$$

from the jaw.

These quantities are assumptions of the model rather than directly measured physical values from the fictional scene.

10.1 Cannonball Momentum

The linear momentum of the cannonball is

$$p = mv. \quad (23)$$

Substituting,

$$p = 15.0 \times 450. \quad (24)$$

Therefore,

$$\boxed{p = 6750 \text{ kg m/s.}} \quad (25)$$

This is the incoming momentum before the collision.

10.2 Momentum Change

The general impulse relationship is

$$J = \Delta p = p_f - p_i. \quad (26)$$

The Python model uses a simplified maximum-transfer assumption.

In the code, the impact force is calculated using

$$F_{\text{impact}} = \frac{m_c v_c}{\Delta t}. \quad (27)$$

This corresponds to taking the relevant momentum change as

$$\Delta p \approx 6750 \text{ kg m/s.} \quad (28)$$

This is an important modeling assumption.

If the cannonball rebounded, the actual momentum change could be larger. If it did not transfer all of its momentum to the sword, the momentum change could be smaller.

Therefore, the calculated impact force should be interpreted as the force associated with the specific momentum-transfer assumption used by the simulation.

10.3 Average Impact Force

Using

$$F_{\text{impact}} = \frac{\Delta p}{\Delta t}, \quad (29)$$

we obtain

$$F_{\text{impact}} = \frac{6750}{0.015}. \quad (30)$$

Therefore,

$$\boxed{F_{\text{impact}} = 450000 \text{ N.}} \quad (31)$$

In kilonewtons,

$$\boxed{F_{\text{impact}} = 450 \text{ kN.}} \quad (32)$$

In tons-force,

$$F_{\text{impact,tf}} = \frac{450000}{9806.65}. \quad (33)$$

Thus,

$$\boxed{F_{\text{impact}} \approx 45.886 \text{ tf.}} \quad (34)$$

This is an **average modeled impact force**, not necessarily the instantaneous peak force.

11 Impact Torque

The impact force is assumed to act at a distance of

$$r_{\text{impact}} = 0.40 \text{ m}$$

from the jaw.

The torque is therefore

$$\tau_{\text{impact}} = F_{\text{impact}} r_{\text{impact}}. \quad (35)$$

Substituting,

$$\tau_{\text{impact}} = 450000 \times 0.40. \quad (36)$$

Therefore,

$$\boxed{\tau_{\text{impact}} = 180000 \text{ N m.}} \quad (37)$$

This is the central torque value used by the simplified model.

12 Theoretical Jaw Force

The jaw is modeled as producing a resisting torque of

$$\tau_{\text{resist}} = F_{\text{jaw}} d_{\text{contact}}. \quad (38)$$

For the limiting condition,

$$\tau_{\text{resist}} = \tau_{\text{impact}}. \quad (39)$$

Therefore,

$$F_{\text{jaw}} d_{\text{contact}} = \tau_{\text{impact}}. \quad (40)$$

Solving for jaw force:

$$F_{\text{jaw}} = \frac{\tau_{\text{impact}}}{d_{\text{contact}}}. \quad (41)$$

Substituting,

$$F_{\text{jaw}} = \frac{180000}{0.05}. \quad (42)$$

Therefore,

$$\boxed{F_{\text{jaw}} = 3600000 \text{ N.}} \quad (43)$$

Converting to tons-force:

$$F_{\text{jaw,tf}} = \frac{3600000}{9806.65}. \quad (44)$$

Therefore,

$$\boxed{F_{\text{jaw}} \approx 367.098 \text{ tf.}} \quad (45)$$

Rounded to two decimal places,

$$\boxed{F_{\text{jaw}} \approx 367.10 \text{ tf.}} \quad (46)$$

Thus, the theoretical threshold produced by this model is

$$\boxed{F_{\text{theoretical}} \approx 367.10 \text{ tf.}} \quad (47)$$

13 Direct Derivation

The entire calculation can be combined into a single equation.

Starting with

$$p = m_c v_c, \quad (48)$$

and

$$F_{\text{impact}} = \frac{p}{\Delta t}, \quad (49)$$

we obtain

$$F_{\text{impact}} = \frac{m_c v_c}{\Delta t}. \quad (50)$$

The impact torque is

$$\tau_{\text{impact}} = \frac{m_c v_c}{\Delta t} r_{\text{impact}}. \quad (51)$$

The required jaw force is then

$$F_{\text{jaw}} = \frac{\tau_{\text{impact}}}{d_{\text{contact}}}. \quad (52)$$

Therefore,

$$\boxed{F_{\text{jaw}} = \frac{m_c v_c r_{\text{impact}}}{\Delta t d_{\text{contact}}}.} \quad (53)$$

Substituting all model constants:

$$F_{\text{jaw}} = \frac{(15.0)(450)(0.40)}{(0.015)(0.05)}. \quad (54)$$

Thus,

$$F_{\text{jaw}} = 3600000 \text{ N}. \quad (55)$$

This gives

$$\boxed{F_{\text{jaw}} \approx 367.10 \text{ tf.}} \quad (56)$$

This direct equation also makes the sensitivity of the result clear.

14 400 tf Comparison Case

The Python simulation also tests a rounded force of

$$400 \text{ tf.}$$

Converting to newtons:

$$F_{400} = 400 \times 9806.65. \quad (57)$$

Therefore,

$$\boxed{F_{400} = 3922660 \text{ N.}} \quad (58)$$

The corresponding resisting torque is

$$\tau_{400} = 3922660 \times 0.05. \quad (59)$$

Therefore,

$$\boxed{\tau_{400} = 196133 \text{ N m.}} \quad (60)$$

The theoretical impact torque is

$$180000 \text{ N m.}$$

Therefore, the additional torque capacity is

$$196133 - 180000 = 16133 \text{ N m.} \quad (61)$$

The relative torque margin is

$$\frac{196133 - 180000}{180000} \times 100. \quad (62)$$

Thus,

$$\boxed{\text{Torque margin} \approx 8.96\%}. \quad (63)$$

Therefore, 400 tf should be described as a rounded comparison case rather than the exact theoretical minimum.

15 Friction Analysis

The Python model also calculates a simplified friction limit.

The Coulomb-friction approximation is

$$F_{\text{friction}} = \mu F_{\text{jaw}}. \quad (64)$$

With

$$\mu = 0.5$$

and

$$F_{\text{jaw}} = 3922660 \text{ N},$$

we obtain

$$F_{\text{friction}} = 0.5 \times 3922660. \quad (65)$$

Therefore,

$$\boxed{F_{\text{friction}} = 1961330 \text{ N}.} \quad (66)$$

The modeled friction limit is greater than the assumed average impact force:

$$1961330 > 450000. \quad (67)$$

The ratio is

$$\frac{1961330}{450000} \approx 4.36. \quad (68)$$

Therefore, under the simplified Coulomb-friction assumption, the friction limit is approximately 4.36 times the modeled average impact force.

However, this does not demonstrate that a real human mouth could secure a sword under such a load.

16 Python Simulation Model

The Python simulation converts the input jaw force from tons-force into newtons:

$$F_{\text{jaw,N}} = F_{\text{jaw,tf}} \times 1000 \times g. \quad (69)$$

The rotational resistance is then calculated as

$$\tau_{\text{resist}} = F_{\text{jaw,N}} d_{\text{contact}}. \quad (70)$$

The code also calculates

$$F_{\text{friction}} = F_{\text{jaw},N}\mu. \quad (71)$$

The gravity torque is calculated as

$$\tau_g = m_s g r_{\text{com}}. \quad (72)$$

These quantities are returned by the simulation for each tested jaw force.

17 Simulation Success Criteria

The simulation does not use a complete rigid-body dynamics solver.

Instead, it uses a piecewise numerical model.

The first condition is:

$$\tau_{\text{resist}} \geq 1.1\tau_{\text{impact}}. \quad (73)$$

When this condition is satisfied, the simulation assigns

$$\text{rotation} = 0^\circ$$

and declares success.

If the resisting torque is between the theoretical impact torque and 1.1 times the impact torque, the simulation calculates a modeled rotation value using its empirical formula.

If the resisting torque is below the impact torque, the simulation uses a different empirical rotation relationship.

Therefore, the simulation should not be described as a full physical rigid-body simulation.

It is more accurately described as a **simplified numerical physics model** based on torque, friction, and predefined response functions.

18 Simulation Results

The following values correspond to the test cases used in the Python program.

Table 2: Python simulation test cases

Case	Jaw Force	Jaw Force (N)	Resistance (N m)	Rotation	Result
1	12 kgf	117.68	5.88	approximately 49°	FAILED
2	80 kgf	784.53	39.23	approximately 49°	FAILED
3	1.6 tf	15690.64	784.53	approximately 48.8°	FAILED
4	10 tf	98066.50	4903.33	approximately 47.7°	FAILED
5	100 tf	980665.00	49033.25	approximately 35.8°	FAILED
6	367.10 tf	approximately 3600000	approximately 180000	approximately 0.50°	SUCCESS
7	400 tf	3922660	196133	0°	SUCCESS
8	1000 tf	9806650	490332.50	0°	SUCCESS
9	2000 tf	19613300	980665	0°	SUCCESS

The exact values near the threshold can differ slightly depending on the binary-search result and floating-point representation.

19 Binary Search for the Minimum

The Python program searches for the minimum successful jaw force using binary search.

The initial interval is

$$0 \leq F_{\text{jaw}} \leq 2000 \text{ tf.} \quad (74)$$

The program repeatedly calculates

$$F_{\text{mid}} = \frac{F_{\text{low}} + F_{\text{high}}}{2}. \quad (75)$$

If the midpoint produces a successful simulation,

$$F_{\text{high}} = F_{\text{mid}}.$$

Otherwise,

$$F_{\text{low}} = F_{\text{mid}}.$$

This process is repeated 30 times.

Therefore, the program is effectively searching for the transition between failure and success.

Because the simulation's success boundary is tied to the torque condition, the binary search converges very close to the analytical threshold.

20 Why the Simulation Gives Approximately 367.10 tf

The theoretical force is

$$F_{\text{theoretical}} = 367.098 \dots \text{ tf.}$$

The corresponding resisting torque is

$$\tau = F_{\text{jaw}} d_{\text{contact}}. \quad (76)$$

At exactly the analytical threshold,

$$\tau \approx 180000 \text{ N m.} \quad (77)$$

Therefore, the simulation threshold is not an independent discovery of a new physical constant.

Rather, the binary-search result is expected to converge toward the same boundary because the simulation itself uses the same impact torque as one of its decision conditions.

This distinction is important when interpreting the result.

The agreement between the analytical calculation and the simulation therefore represents **internal consistency of the model**, rather than independent experimental validation.

21 Comparison With Real Animals

The modeled value can be compared with approximate bite-force estimates reported or modeled for various animals.

These values should be treated as approximate because animal bite force depends on jaw position, tooth position, body size, measurement method, and whether the number is experimentally measured or theoretically estimated.

Table 3: Approximate bite-force comparison

Subject	Approximate Bite Force	400 tf / Bite Force
Megalodon	10.20–18.36 tf	21.8–39.2×
<i>Tyrannosaurus rex</i>	3.57–5.81 tf	68.8–112.0×
Great white shark	1.84 tf	217.4×
Orca	0.92–1.84 tf	217.4–434.8×
Tiger	0.41–0.61 tf	655.7–975.6×
Lion	0.41–0.51 tf	784.3–975.6×
Average adult human	0.05–0.09 tf	4444–8000×

These comparisons are included primarily to illustrate the magnitude of the modeled result.

They should not be interpreted as a rigorous biological comparison because the measurement conditions are not necessarily equivalent.

22 Megalodon Comparison

Using the upper estimate of

$$18.36 \text{ tf},$$

the ratio is

$$\frac{400}{18.36} = 21.786 \dots \quad (78)$$

or approximately

$$\boxed{21.8 \times .} \quad (79)$$

Using the lower estimate of

$$10.20 \text{ tf},$$

we obtain

$$\frac{400}{10.20} = 39.216 \dots \quad (80)$$

Therefore,

$$\boxed{400 \text{ tf} \approx 21.8\text{--}39.2 \times \text{the estimated megalodon bite force.}} \quad (81)$$

Using the modeled minimum of 367.10 tf instead gives

$$\frac{367.10}{18.36} \approx 20.0 \quad (82)$$

and

$$\frac{367.10}{10.20} \approx 36.0. \quad (83)$$

Thus,

$$\boxed{367.10 \text{ tf} \approx 20.0\text{--}36.0 \times \text{the estimated megalodon bite force.}} \quad (84)$$

23 Young Zoro Comparison

For a fictional comparison, the model uses an assumed Young Zoro jaw-force value of

$$1.6 \text{ tf.}$$

This value is a comparison assumption and is not presented as a canonically measured value.

The ratio between the modeled threshold and this assumed value is

$$\frac{367.10}{1.6} = 229.4375. \quad (85)$$

Therefore,

$$\boxed{F_{\text{modeled}} \approx 229.4 \times F_{\text{Young}}.} \quad (86)$$

For the rounded 400 tf case,

$$\frac{400}{1.6} = 250. \quad (87)$$

Thus,

$$\boxed{400 \text{ tf} = 250 \times 1.6 \text{ tf.}} \quad (88)$$

24 Sensitivity Analysis

The final value of 367.10 tf depends strongly on several assumptions.

The direct equation is

$$F_{\text{jaw}} = \frac{m_c v_c r_{\text{impact}}}{\Delta t d_{\text{contact}}}. \quad (89)$$

This equation allows the dependence on each parameter to be examined.

24.1 Effect of Collision Time

The jaw force is inversely proportional to collision time:

$$F_{\text{jaw}} \propto \frac{1}{\Delta t}. \quad (90)$$

Therefore, if the collision time doubles, the predicted jaw force is approximately halved.

For example, changing

$$\Delta t = 0.015 \text{ s}$$

to

$$\Delta t = 0.030 \text{ s}$$

would reduce the theoretical result from approximately

$$367.10 \text{ tf}$$

to approximately

$$183.55 \text{ tf}.$$

Conversely, reducing the collision time would increase the predicted force.

This makes collision duration one of the most influential assumptions in the model.

24.2 Effect of Impact Distance

The jaw force is directly proportional to the impact distance:

$$F_{\text{jaw}} \propto r_{\text{impact}}. \quad (91)$$

If the impact point were twice as far from the jaw, the required jaw force would also be approximately twice as large.

For example,

$$r_{\text{impact}} = 0.80 \text{ m}$$

would produce approximately

$$734.20 \text{ tf}$$

under otherwise identical assumptions.

Similarly, reducing the impact distance would reduce the required jaw force.

24.3 Effect of Jaw Contact Distance

The jaw force is inversely proportional to the effective contact distance:

$$F_{\text{jaw}} \propto \frac{1}{d_{\text{contact}}}. \quad (92)$$

Therefore, increasing the effective contact distance would reduce the required jaw force.

For example, doubling

$$d_{\text{contact}}$$

from 0.05 m to 0.10 m would reduce the theoretical force by approximately half.

This demonstrates that the assumed jaw geometry is also a major source of uncertainty.

25 Important Distinction: Average Force vs. Peak Force

The calculation above produces an average impact force:

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t}. \quad (93)$$

It does **not** directly calculate the peak instantaneous force.

The peak force would depend on the detailed force–time curve during the collision.

For example, if the force changes with time,

$$F = F(t), \quad (94)$$

then the impulse is

$$J = \int F(t) dt. \quad (95)$$

The average force is

$$F_{\text{avg}} = \frac{1}{\Delta t} \int F(t) dt. \quad (96)$$

Therefore, a statement such as

“The cannonball produces a peak force of 450000 N.”

would not be justified by this calculation.

The correct statement is:

The model estimates an average impact force of 450000 N over the assumed 0.015 s collision interval.

This distinction is important for the physical interpretation of the result.

26 Peak Force Considerations

The current model does not contain enough information to calculate a unique peak impact force.

To calculate peak force physically, additional information would be needed, such as:

- the actual force–time profile,
- collision stiffness,
- deformation of the cannonball,
- deformation of the sword,
- coefficient of restitution,
- exact impact angle,
- contact area,
- relative velocity during contact,
- and the effective masses of the interacting bodies.

Therefore, the value

450000 N

should not be labeled as a measured peak force.

It is the average force implied by the assumed momentum transfer and collision duration.

Consequently, the 367.10 tf result should also be described as the jaw force required to balance the **modeled average impact torque**, rather than a guaranteed biological requirement against an unknown instantaneous peak force.

27 Limitations

This model has several important limitations.

27.1 Cannonball Parameters

The cannonball mass, velocity, and collision duration are assumptions.

They are not directly measured physical quantities obtained from the fictional scene. Changing these values changes the final result.

27.2 Momentum Transfer

The calculation effectively assumes that the relevant incoming momentum is transferred over the collision interval.

A real collision may involve:

- rebound,
- partial momentum transfer,
- deformation,
- fragmentation,
- rotation,
- and motion of the sword and target.

Therefore, the actual momentum change could differ.

27.3 Rigid Sword Assumption

The sword is represented by a simplified rigid-body model.

Real swords have distributed mass and finite stiffness.

Their response may involve bending, vibration, and deformation.

These effects are not represented in the simple torque equation.

27.4 Jaw Geometry

The model represents the complex interaction between teeth, jaw, muscles, and sword using a single effective distance:

$$d_{\text{contact}} = 0.05 \text{ m.}$$

This is a substantial simplification.

Actual human jaw mechanics cannot be accurately represented by a single force multiplied by a single lever arm.

27.5 Friction

The friction model uses a constant coefficient

$$\mu = 0.5.$$

Real tooth–metal contact would depend on surface condition, contact pressure, moisture, geometry, and deformation.

Therefore, the friction result is only a simplified estimate.

27.6 Simulation Rotation

The Python program does not numerically integrate the complete equations of rigid-body rotational dynamics.

Instead, it uses predefined empirical rotation formulas.

Therefore, the reported rotation values such as

$$0.50^\circ$$

are outputs of the simulation’s response function rather than experimentally validated physical predictions.

28 Analytical Result vs. Simulation Result

The analytical model gives

$$\boxed{F_{\text{theoretical}} \approx 367.10 \text{ tf.}} \quad (97)$$

The Python binary search also converges to approximately the same threshold.

This agreement is expected because the simulation uses the same impact torque and contact-distance relationship in its success condition.

Therefore, the correct interpretation is:

The analytical calculation and Python simulation are internally consistent with one another.

However, this does not constitute independent experimental validation.

Both methods are based on the same assumptions.

29 Final Result

The model begins with the cannonball momentum:

$$p = m_c v_c = 15.0 \times 450 = 6750 \text{ kg m/s.} \quad (98)$$

Using the assumed collision duration,

$$F_{\text{impact}} = \frac{6750}{0.015} = 450000 \text{ N.} \quad (99)$$

At an impact distance of

$$0.40 \text{ m,}$$

the corresponding torque is

$$\tau_{\text{impact}} = 450000 \times 0.40 = 180000 \text{ N m.} \quad (100)$$

The required jaw force is therefore

$$F_{\text{jaw}} = \frac{180000}{0.05} = 3600000 \text{ N.} \quad (101)$$

Converting to tons-force:

$$F_{\text{jaw}} = \frac{3600000}{9806.65} \approx 367.10 \text{ tf.} \quad (102)$$

Thus,

$$\boxed{\text{Modeled Zoro Jaw Force} \approx 367.10 \text{ tf}} \quad (103)$$

for the specific cannonball scenario used in this research.

30 Conclusion

This study constructed a simplified mechanical model of Zoro's Three-Sword Style.

The model assumes a 1.5 kg sword with a length of 1.35 m and an effective jaw contact distance of 0.05 m.

A 15.0 kg cannonball traveling at 450 m/s was modeled with an assumed collision duration of 0.015 s.

Under the maximum-momentum-transfer assumption, the resulting average impact force is

$$450000 \text{ N.}$$

At a lever arm of 0.40 m, this corresponds to

$$180000 \text{ N m}$$

of impact torque.

Balancing this torque using the assumed jaw contact distance gives

$$3600000 \text{ N}$$

or approximately

$$367.10 \text{ tf.}$$

The Python simulation produces a minimum-success threshold of approximately the same magnitude because its success condition is constructed around the same torque relationship.

The 400 tf test case provides approximately a

$$8.96\%$$

torque margin over the theoretical threshold.

However, the result should not be interpreted as Zoro's actual biological bite force.

The calculation depends strongly on assumptions concerning collision time, momentum transfer, impact distance, jaw contact geometry, friction, and the fictional cannonball scenario.

Most importantly, the calculated 450000 N is an **average impact force**, not a measured peak force.

Therefore, the scientifically appropriate conclusion is:

*Under the assumptions of this simplified mechanical model, a jaw clamping force of approximately **367.10 tf** would be required to balance the modeled average impact torque of the selected cannonball scenario.*

The result is best interpreted as a physics-based estimate for a fictional scenario rather than as a real-world measurement.

Zoro's mouth is basically doing the job of a hydraulic press.

Summary of Key Values

Table 4: Summary of the main calculated quantities

Quantity	Value
Sword mass	1.5 kg
Sword length	1.35 m
Center-of-mass distance	0.40 m
Jaw contact distance	0.05 m
Cannonball mass	15.0 kg
Cannonball velocity	450 m/s
Collision duration	0.015 s
Cannonball momentum	6750 kg m/s
Average impact force	450000 N
Average impact force	450 kN
Average impact force	45.89 tf
Impact torque	180000 N m
Theoretical jaw force	3600000 N
Theoretical jaw force	367.10 tf
400 tf torque	196133 N m
400 tf torque margin	8.96%